

### For an Optimal Portfolio Design Point, 70-trading Days Forward Return for Test Data

| Obs | r1      | r2      | r3      | r4      | r5      | r6      | r7      | r8      | y <sub>i</sub> | $\hat{y}$ | (y <sub>i</sub> - $\hat{y}$ ) <sup>2</sup> | L( $\delta_i$ ) |
|-----|---------|---------|---------|---------|---------|---------|---------|---------|----------------|-----------|--------------------------------------------|-----------------|
| 1   | 9.2776  | 0.0014  | 8.0362  | 1.9791  | 0.2489  | 7.5212  | 5.7055  | 23.7722 | 7.5084         | 3.12223   | 19.238                                     | 3.40            |
| 2   | 9.3785  | -1.7980 | 8.7131  | 1.7493  | -1.6732 | 7.2958  | 5.5459  | 25.6672 | 7.8436         | 3.12223   | 22.291                                     | 3.73            |
| 3   | 9.0382  | -2.0940 | 7.9477  | 0.6365  | -2.0136 | 6.6989  | 4.6793  | 21.9460 | 6.7612         | 3.12223   | 13.242                                     | 2.67            |
| 4   | 8.1246  | -1.2769 | 7.5625  | 0.8525  | -1.6409 | 5.2519  | 4.5453  | 15.4050 | 5.9590         | 3.12223   | 8.048                                      | 1.90            |
| 5   | 7.3831  | -1.9593 | 7.2745  | -0.1081 | -2.0548 | 5.7370  | 4.2049  | 17.2203 | 5.6602         | 3.12223   | 6.441                                      | 1.62            |
| 6   | 7.1056  | 1.0791  | 6.3933  | 0.8134  | 0.6803  | 5.6219  | 4.1561  | 16.7585 | 5.5144         | 3.12223   | 5.722                                      | 1.48            |
| 7   | 7.3097  | -0.1678 | 6.7432  | 0.8436  | -0.4517 | 4.3722  | 4.2377  | 15.6142 | 5.5251         | 3.12223   | 5.774                                      | 1.49            |
| 8   | 7.8725  | 2.4977  | 8.0237  | 0.8920  | 1.9811  | 4.8715  | 4.6032  | 16.9889 | 6.3082         | 3.12223   | 10.150                                     | 2.23            |
| 9   | 7.1874  | 3.3045  | 7.2611  | 0.3132  | 2.5058  | 5.2357  | 4.0111  | 12.9062 | 5.3929         | 3.12223   | 5.156                                      | 1.37            |
| 10  | 7.5444  | 0.0488  | 7.4394  | 1.0944  | -0.3322 | 4.5038  | 4.8328  | 12.4181 | 5.6879         | 3.12223   | 6.582                                      | 1.64            |
| 11  | 7.5706  | -0.1402 | 8.3146  | 2.4023  | -1.1272 | 3.9115  | 5.4804  | 12.1569 | 6.4159         | 3.12223   | 10.848                                     | 2.33            |
| 12  | 8.2443  | 1.3350  | 8.2339  | 1.9015  | 0.1581  | 4.3205  | 5.6121  | 14.4058 | 6.5470         | 3.12223   | 11.729                                     | 2.46            |
| 13  | 8.3523  | 1.3190  | 8.2720  | 0.7347  | 0.2213  | 3.8799  | 4.8602  | 14.9788 | 6.2049         | 3.12223   | 9.503                                      | 2.13            |
| 14  | 9.0338  | 2.7582  | 10.2973 | 0.7601  | 0.8919  | 3.7441  | 5.3861  | 9.9971  | 6.7146         | 3.12223   | 12.905                                     | 2.62            |
| 15  | 8.7191  | 0.6085  | 10.0924 | -1.3741 | -1.1634 | 2.2609  | 4.3500  | 10.8584 | 5.9111         | 3.12223   | 7.778                                      | 1.85            |
| 16  | 9.0577  | 1.5388  | 10.9615 | -1.7594 | -0.1854 | 2.9090  | 4.1935  | 12.1805 | 6.3168         | 3.12223   | 10.205                                     | 2.24            |
| 17  | 9.8406  | 1.8694  | 11.0320 | -1.4332 | 0.2045  | 3.1691  | 4.5433  | 10.9523 | 6.4407         | 3.12223   | 11.012                                     | 2.35            |
| 18  | 8.6142  | 0.6608  | 9.0223  | 0.5383  | -0.7525 | 3.0483  | 4.8342  | 10.9598 | 6.0799         | 3.12223   | 8.748                                      | 2.01            |
| 19  | 7.7779  | -0.3021 | 6.5495  | -0.1235 | -1.5486 | 2.1775  | 4.1920  | 13.5327 | 4.9509         | 3.12223   | 3.344                                      | 0.99            |
| 20  | 6.4335  | -0.8342 | 5.1318  | -0.4220 | -2.0751 | 1.7822  | 3.6764  | 15.1605 | 4.2581         | 3.12223   | 1.290                                      | 0.46            |
| 21  | 5.0402  | -2.1144 | 3.3426  | -1.6361 | -3.0423 | 1.1170  | 2.1881  | 12.0090 | 2.6195         | 3.12223   | 0.253                                      | 0.15            |
| 22  | 3.2855  | -3.3653 | 3.4198  | 1.6325  | -3.8798 | 2.0587  | 3.5831  | 7.9378  | 3.2148         | 3.12223   | 0.009                                      | 0.00            |
| 23  | 3.5812  | -3.4740 | 4.7836  | 3.8132  | -3.8580 | 3.7132  | 5.0186  | 10.4624 | 4.8171         | 3.12223   | 2.873                                      | 0.88            |
| 24  | 5.1271  | -4.5297 | 6.0746  | 4.4429  | -4.5941 | 4.0240  | 5.9312  | 12.3394 | 5.8937         | 3.12223   | 7.681                                      | 1.83            |
| 25  | 4.8452  | -3.7159 | 5.8115  | 3.3415  | -4.3142 | 4.0410  | 4.9399  | 12.6565 | 5.3943         | 3.12223   | 5.162                                      | 1.38            |
| 26  | 4.5185  | -2.4817 | 4.8859  | 3.2556  | -3.1354 | 2.4227  | 4.6908  | 10.9162 | 4.7444         | 3.12223   | 2.631                                      | 0.82            |
| 27  | 4.7659  | -1.7817 | 5.8220  | 4.6028  | -2.8283 | 2.4924  | 5.2192  | 7.3576  | 5.2595         | 3.12223   | 4.568                                      | 1.26            |
| 28  | 5.1543  | -2.3009 | 5.7961  | 6.5661  | -3.0918 | 4.2055  | 6.1635  | 10.7694 | 6.2714         | 3.12223   | 9.917                                      | 2.19            |
| 29  | 3.7529  | -3.5322 | 3.9433  | 5.2873  | -4.0827 | 3.4048  | 4.8946  | 8.5831  | 4.6943         | 3.12223   | 2.471                                      | 0.78            |
| 30  | 3.4122  | -0.8903 | 3.6337  | 5.6308  | -1.8917 | 3.3859  | 5.0999  | 9.7163  | 4.7870         | 3.12223   | 2.771                                      | 0.85            |
| 31  | 5.7275  | -0.9857 | 7.6290  | 9.9609  | -1.7701 | 2.9962  | 8.5749  | 4.5661  | 7.6633         | 3.12223   | 20.621                                     | 3.55            |
| 32  | 7.3446  | -0.0311 | 8.2459  | 9.0461  | -1.1412 | 3.8703  | 8.2541  | 4.0159  | 7.7527         | 3.12223   | 21.441                                     | 3.64            |
| 33  | 9.0659  | 3.0145  | 10.8941 | 12.3457 | 0.4168  | 4.8617  | 10.4676 | 5.4848  | 10.2863        | 3.12223   | 51.324                                     | 6.16            |
| 34  | 11.9388 | 6.8766  | 16.0636 | 14.9477 | 3.3866  | 7.3060  | 12.9132 | 1.3559  | 13.3416        | 3.12223   | 104.435                                    | 9.22            |
| 35  | 13.9951 | 9.8418  | 19.0141 | 20.1392 | 6.0140  | 10.5424 | 16.4246 | 7.2214  | 17.1497        | 3.12223   | 196.770                                    | 13.03           |

| Obs | r1      | r2      | r3      | r4      | r5      | r6      | r7      | r8       | y <sub>i</sub> | $\hat{y}$ | $(y_i - \hat{y})^2$ | L( $\delta_i$ ) |
|-----|---------|---------|---------|---------|---------|---------|---------|----------|----------------|-----------|---------------------|-----------------|
| 36  | 13.4366 | 7.7788  | 16.6864 | 17.3383 | 4.2468  | 7.5144  | 14.8713 | 3.4617   | 14.7752        | 3.12223   | 135.792             | 10.65           |
| 37  | 12.8405 | 8.1831  | 15.4493 | 16.8782 | 4.7547  | 6.3953  | 14.3727 | 3.9073   | 14.0514        | 3.12223   | 119.448             | 9.93            |
| 38  | 11.3754 | 7.7991  | 13.2502 | 13.0810 | 4.0370  | 5.5888  | 11.8299 | 3.7693   | 11.6990        | 3.12223   | 73.560              | 7.58            |
| 39  | 10.3673 | 6.6086  | 12.4802 | 11.3984 | 3.0105  | 3.5534  | 10.6608 | 4.8982   | 10.7321        | 3.12223   | 57.911              | 6.61            |
| 40  | 9.3936  | 5.7087  | 11.9680 | 10.7938 | 2.2084  | 3.2086  | 9.9715  | 4.9051   | 10.1862        | 3.12223   | 49.900              | 6.06            |
| 41  | 9.3481  | 7.0217  | 12.9206 | 8.9662  | 2.9359  | 2.8323  | 8.9935  | 3.3736   | 9.8303         | 3.12223   | 44.998              | 5.71            |
| 42  | 9.0806  | 7.8031  | 13.2744 | 8.4923  | 3.2016  | 1.8769  | 8.7031  | 3.7857   | 9.7982         | 3.12223   | 44.569              | 5.68            |
| 43  | 8.8741  | 7.8483  | 13.1727 | 7.3585  | 3.3521  | 2.0438  | 8.0571  | 4.3289   | 9.4187         | 3.12223   | 39.645              | 5.30            |
| 44  | 9.7196  | 5.8618  | 15.0493 | 6.8931  | 1.7044  | 0.9052  | 7.9408  | -1.2417  | 9.5469         | 3.12223   | 41.276              | 5.43            |
| 45  | 9.2347  | 4.8094  | 13.7058 | 7.1344  | 0.4047  | -0.2315 | 7.9544  | -3.1240  | 8.8148         | 3.12223   | 32.406              | 4.70            |
| 46  | 9.6514  | 7.4431  | 13.9880 | 4.9014  | 2.3984  | -0.0229 | 7.1501  | -4.4835  | 8.1914         | 3.12223   | 25.696              | 4.08            |
| 47  | 9.8718  | 7.6975  | 13.9944 | 5.3900  | 2.6283  | -0.0157 | 7.3756  | -3.5555  | 8.4516         | 3.12223   | 28.402              | 4.33            |
| 48  | 10.1657 | 6.2424  | 13.2251 | 7.7693  | 1.4509  | 0.9571  | 8.5306  | -3.9147  | 8.9245         | 3.12223   | 33.666              | 4.81            |
| 49  | 9.1370  | 6.2706  | 12.2720 | 7.9644  | 1.8445  | 1.6000  | 8.0254  | -2.9721  | 8.5631         | 3.12223   | 29.603              | 4.45            |
| 50  | 8.8952  | 2.0180  | 11.1694 | 9.7930  | -1.6267 | 0.9700  | 8.1880  | -2.9793  | 8.5342         | 3.12223   | 29.289              | 4.42            |
| 51  | 8.8489  | 2.6866  | 11.5540 | 10.5423 | -1.4338 | 1.0488  | 8.5711  | -3.0639  | 8.9319         | 3.12223   | 33.753              | 4.81            |
| 52  | 9.1919  | 2.0749  | 12.0857 | 11.5919 | -1.9462 | 1.5378  | 9.3335  | -2.5631  | 9.5963         | 3.12223   | 41.914              | 5.48            |
| 53  | 8.0221  | 1.7621  | 10.9353 | 10.9903 | -1.9672 | 1.3261  | 8.3264  | -1.3742  | 8.8580         | 3.12223   | 32.899              | 4.74            |
| 54  | 7.3903  | 0.7781  | 10.0318 | 10.2793 | -2.2747 | 0.3453  | 8.0279  | -0.1213  | 8.2825         | 3.12223   | 26.628              | 4.17            |
| 55  | 8.2340  | 2.3933  | 12.3677 | 14.1183 | -0.5081 | 2.0929  | 10.0979 | 4.8013   | 11.0655        | 3.12223   | 63.096              | 6.94            |
| 56  | 7.4983  | 3.1084  | 12.2753 | 14.8224 | 0.0720  | 2.0533  | 9.9496  | 4.3172   | 11.1055        | 3.12223   | 63.733              | 6.98            |
| 57  | 8.0737  | 2.9893  | 12.0558 | 15.0970 | -0.4029 | 1.7669  | 10.1309 | -0.0570  | 10.7621        | 3.12223   | 58.367              | 6.64            |
| 58  | 8.4113  | 2.4598  | 13.5460 | 18.9054 | -0.7197 | 3.4114  | 12.1519 | 2.2891   | 12.8993        | 3.12223   | 95.591              | 8.78            |
| 59  | 6.9553  | 0.4380  | 11.8555 | 18.6893 | -2.3548 | 2.4639  | 11.0535 | -0.1690  | 11.6475        | 3.12223   | 72.680              | 7.53            |
| 60  | 4.6578  | 0.6272  | 8.3463  | 13.8722 | -2.0120 | 0.2725  | 7.8297  | -3.9550  | 7.9881         | 3.12223   | 23.677              | 3.87            |
| 61  | 5.8537  | 1.0977  | 9.0517  | 13.8244 | -1.6462 | 0.0467  | 8.4911  | -5.3406  | 8.3044         | 3.12223   | 26.855              | 4.19            |
| 62  | 3.6414  | -0.3903 | 6.3387  | 9.1143  | -3.1543 | -3.6513 | 5.3618  | -8.1863  | 5.0140         | 3.12223   | 3.579               | 1.04            |
| 63  | 3.3752  | -1.2047 | 5.4289  | 9.4899  | -3.9370 | -3.2475 | 5.3720  | -13.2627 | 4.2932         | 3.12223   | 1.371               | 0.48            |
| 64  | 3.4671  | -0.6309 | 5.2365  | 11.7382 | -3.4858 | -2.2575 | 6.7251  | -12.3601 | 5.1037         | 3.12223   | 3.926               | 1.12            |
| 65  | 3.9652  | -1.2391 | 4.8437  | 11.9382 | -3.9700 | -1.3290 | 7.0060  | -11.3800 | 5.1860         | 3.12223   | 4.259               | 1.19            |
| 66  | 3.1667  | -2.4291 | 3.7518  | 10.1464 | -4.5682 | -1.8961 | 5.6610  | -11.9695 | 3.9651         | 3.12223   | 0.710               | 0.27            |
| 67  | 1.0313  | -3.1525 | 1.1002  | 6.7060  | -4.9553 | -2.6346 | 3.2136  | -13.8310 | 1.3341         | 3.12223   | 3.197               | 3.19            |
| 68  | 1.3265  | -2.2954 | 2.1269  | 6.8599  | -4.2331 | -3.5130 | 3.4253  | -14.6452 | 1.7389         | 3.12223   | 1.914               | 1.60            |
| 69  | 0.3870  | -2.0017 | 1.4345  | 5.0646  | -3.9138 | -3.6900 | 2.1968  | -12.7791 | 0.9258         | 3.12223   | 4.824               | 5.80            |
| 70  | 0.9038  | -1.7750 | 1.5654  | 6.7692  | -3.6158 | -3.6442 | 3.3281  | -11.6976 | 1.7033         | 3.12223   | 2.013               | 1.71            |
| 71  | 1.7951  | 0.6431  | 4.1060  | 7.7370  | -1.0727 | -3.4377 | 4.3408  | -14.5735 | 2.9531         | 3.12223   | 0.029               | 0.02            |
| 72  | 0.8195  | 2.9399  | 2.2822  | 5.6799  | 1.1890  | -2.9833 | 2.9657  | -14.7180 | 1.4689         | 3.12223   | 2.734               | 2.57            |

| Obs | r1      | r2       | r3      | r4      | r5      | r6      | r7      | r8       | y <sub>i</sub> | $\hat{y}$ | (y <sub>i</sub> - $\hat{y}$ ) <sup>2</sup> | L( $\delta_i$ ) |
|-----|---------|----------|---------|---------|---------|---------|---------|----------|----------------|-----------|--------------------------------------------|-----------------|
| 73  | 0.2364  | 2.7176   | 1.2737  | 5.1490  | 0.9792  | -3.7746 | 2.5743  | -12.0668 | 1.0295         | 3.12223   | 4.380                                      | 5.01            |
| 74  | -0.1235 | 2.0808   | 1.4241  | 4.3617  | 0.5790  | -3.3753 | 1.8278  | -7.7676  | 1.1485         | 3.12223   | 3.895                                      | 4.22            |
| 75  | -0.2402 | 2.2417   | 1.3737  | 4.6684  | 0.7932  | -4.3268 | 1.6199  | -10.9915 | 0.8600         | 3.12223   | 5.118                                      | 6.34            |
| 76  | 0.4485  | 0.8183   | 3.3004  | 5.0523  | -1.1031 | -4.4059 | 2.3162  | -10.7937 | 1.8674         | 3.12223   | 1.575                                      | 1.25            |
| 77  | 2.5807  | 1.6306   | 5.7211  | 6.7860  | -0.3906 | -3.2733 | 4.0131  | -7.6639  | 3.9908         | 3.12223   | 0.754                                      | 0.29            |
| 78  | 2.6438  | 0.6458   | 4.7172  | 6.3459  | -1.4387 | -2.0013 | 3.9961  | -7.1596  | 3.5564         | 3.12223   | 0.189                                      | 0.08            |
| 79  | 3.5399  | 0.3466   | 6.8620  | 7.9522  | -1.9765 | -2.3970 | 5.1274  | -6.5799  | 5.0776         | 3.12223   | 3.824                                      | 1.10            |
| 80  | 3.9497  | 3.0263   | 7.3086  | 7.5347  | 0.3737  | -1.5901 | 5.1154  | -2.3270  | 5.6092         | 3.12223   | 6.185                                      | 1.57            |
| 81  | 4.8788  | 1.8891   | 7.6342  | 8.5215  | -0.1736 | 0.3975  | 5.7634  | -1.9391  | 6.2499         | 3.12223   | 9.783                                      | 2.17            |
| 82  | 4.2675  | 0.4526   | 7.2297  | 7.9775  | -1.3288 | 0.1001  | 5.3759  | -5.1641  | 5.5409         | 3.12223   | 5.850                                      | 1.51            |
| 83  | 4.2926  | -0.6581  | 7.1561  | 7.7791  | -2.4679 | 0.3645  | 5.4311  | -5.5063  | 5.4347         | 3.12223   | 5.347                                      | 1.41            |
| 84  | 3.0888  | -2.1858  | 3.7842  | 3.0451  | -3.5700 | -2.1438 | 2.2875  | -7.0564  | 2.1102         | 3.12223   | 1.024                                      | 0.74            |
| 85  | 4.4933  | -0.4085  | 3.8105  | 4.7779  | -1.8836 | -0.5082 | 3.2759  | -6.7602  | 2.9171         | 3.12223   | 0.042                                      | 0.02            |
| 86  | 4.4194  | -0.3442  | 3.2796  | 6.0642  | -1.8692 | 0.0911  | 3.5229  | -8.0121  | 2.9846         | 3.12223   | 0.019                                      | 0.01            |
| 87  | 3.3137  | -0.1170  | 4.2090  | 5.5595  | -1.5362 | -0.0194 | 3.2055  | -5.7996  | 3.2913         | 3.12223   | 0.029                                      | 0.01            |
| 88  | 3.4177  | -1.9735  | 3.5810  | 5.6835  | -2.8844 | -0.3729 | 3.1614  | -6.7868  | 2.9545         | 3.12223   | 0.028                                      | 0.01            |
| 89  | 2.3357  | -2.9152  | 2.0327  | 4.0178  | -4.3572 | -2.3781 | 1.7020  | -10.1273 | 1.2538         | 3.12223   | 3.491                                      | 3.61            |
| 90  | 3.7695  | -2.1318  | 4.3186  | 5.3880  | -3.2414 | -1.0250 | 2.9152  | -9.4780  | 2.9081         | 3.12223   | 0.046                                      | 0.02            |
| 91  | 4.8703  | 0.7850   | 5.5459  | 7.3927  | -0.1828 | 1.1805  | 4.2456  | -5.8378  | 4.6129         | 3.12223   | 2.222                                      | 0.72            |
| 92  | 5.5780  | -0.8938  | 5.1330  | 7.4632  | -1.6478 | 2.5104  | 4.5275  | -3.3965  | 4.8001         | 3.12223   | 2.815                                      | 0.86            |
| 93  | 5.2992  | -0.8034  | 5.0623  | 5.6254  | -1.8119 | 1.5782  | 3.7928  | -2.7315  | 4.1972         | 3.12223   | 1.156                                      | 0.42            |
| 94  | 4.8194  | 1.1156   | 4.3792  | 6.3916  | 0.0510  | 1.8518  | 3.8083  | -0.4015  | 4.3275         | 3.12223   | 1.453                                      | 0.50            |
| 95  | 5.0931  | 3.3091   | 4.0172  | 6.5581  | 2.3207  | 2.2460  | 4.1393  | -2.0156  | 4.1865         | 3.12223   | 1.133                                      | 0.41            |
| 96  | 4.6580  | 0.9501   | 2.3901  | 5.2773  | 0.6539  | 2.9933  | 3.2991  | -1.6990  | 3.0914         | 3.12223   | 0.001                                      | 0.00            |
| 97  | 6.3830  | 0.2013   | 4.2571  | 9.4454  | 0.2238  | 4.7361  | 6.1038  | 2.9898   | 5.8808         | 3.12223   | 7.610                                      | 1.82            |
| 98  | 6.2793  | 1.9405   | 5.1663  | 10.3221 | 1.9479  | 4.9910  | 6.7165  | 1.3987   | 6.4337         | 3.12223   | 10.966                                     | 2.35            |
| 99  | 6.4158  | 1.8301   | 5.4463  | 10.7094 | 1.6364  | 5.1547  | 7.0745  | 2.1394   | 6.7711         | 3.12223   | 13.314                                     | 2.67            |
| 100 | 6.0950  | 1.4223   | 5.1649  | 9.9927  | 1.3031  | 4.8230  | 6.7991  | 1.1314   | 6.3048         | 3.12223   | 10.129                                     | 2.22            |
| 101 | 2.5665  | 0.7884   | -0.2340 | 6.5871  | 0.5818  | 6.0323  | 3.7460  | 7.1385   | 3.1650         | 3.12223   | 0.002                                      | 0.00            |
| 102 | -0.6704 | -2.2223  | -3.8347 | 3.3534  | -2.1907 | 4.1428  | 1.0316  | 1.8254   | -0.2813        | 3.12223   | 11.584                                     | 25.67           |
| 103 | -0.8922 | -3.0606  | -4.8431 | 1.3854  | -2.1378 | 3.1942  | 0.0179  | 1.3649   | -1.4314        | 3.12223   | 20.735                                     | 89.42           |
| 104 | -1.7583 | -5.3845  | -6.6995 | -0.2996 | -3.8992 | 3.4660  | -1.3062 | 6.3372   | -2.4286        | 3.12223   | 30.812                                     | 250.91          |
| 105 | -1.2950 | -5.7538  | -4.4710 | 0.3145  | -3.7281 | 5.3523  | -0.4596 | 9.1772   | -0.9180        | 3.12223   | 16.324                                     | 51.80           |
| 106 | -1.3814 | -4.6495  | -5.6559 | -1.9783 | -2.9303 | 4.8829  | -1.7957 | 7.9456   | -2.2800        | 3.12223   | 29.184                                     | 215.49          |
| 107 | -1.3819 | -6.1018  | -5.0844 | -2.1883 | -5.0264 | 3.4437  | -1.7606 | 6.3695   | -2.3138        | 3.12223   | 29.551                                     | 223.10          |
| 108 | -0.9945 | -8.2026  | -5.3636 | -0.1751 | -6.9376 | 3.9910  | -0.7193 | 8.1126   | -1.5887        | 3.12223   | 22.193                                     | 105.44          |
| 109 | -2.1937 | -10.0252 | -7.7876 | -1.7968 | -8.7158 | 2.5411  | -2.2735 | 4.9459   | -3.6218        | 3.12223   | 45.481                                     | 841.20          |

| Obs | r1       | r2       | r3       | r4       | r5       | r6       | r7       | r8      | y <sub>i</sub> | $\hat{y}$ | (y <sub>i</sub> - $\hat{y}$ ) <sup>2</sup> | L( $\delta_i$ ) |
|-----|----------|----------|----------|----------|----------|----------|----------|---------|----------------|-----------|--------------------------------------------|-----------------|
| 110 | -1.2113  | -11.5438 | -6.4247  | -1.1787  | -9.4916  | 3.5925   | -1.9289  | 4.5633  | -2.8025        | 3.12223   | 35.102                                     | 367.25          |
| 111 | -1.6095  | -11.2984 | -8.2370  | -2.1271  | -9.3060  | 4.5520   | -2.5411  | 6.2152  | -3.7000        | 3.12223   | 46.543                                     | 910.21          |
| 112 | -2.6991  | -10.8424 | -10.4253 | -3.4251  | -8.9755  | 3.3688   | -3.7534  | 5.6296  | -5.2423        | 3.12223   | 69.965                                     | 4282.74         |
| 113 | -5.1485  | -10.2780 | -12.8057 | -6.8379  | -8.5738  | 0.4464   | -6.3983  | 1.7696  | -8.0504        | 3.12223   | 124.827                                    | 71141.46        |
| 114 | -7.2324  | -9.0983  | -14.8529 | -7.3957  | -7.5551  | 0.2483   | -7.3023  | 7.2836  | -8.8014        | 3.12223   | 142.174                                    | 150782.67       |
| 115 | -7.6900  | -8.4647  | -14.0539 | -6.9946  | -6.6062  | 1.7214   | -7.4459  | 9.5206  | -8.1660        | 3.12223   | 127.425                                    | 79866.68        |
| 116 | -9.0142  | -9.1847  | -15.3186 | -7.0959  | -6.8404  | 1.3312   | -8.0239  | 10.1299 | -8.8409        | 3.12223   | 143.117                                    | 156852.26       |
| 117 | -8.5579  | -8.6813  | -14.4302 | -6.3412  | -6.3723  | 1.3040   | -7.3306  | 12.3778 | -7.9664        | 3.12223   | 122.957                                    | 65409.46        |
| 118 | -7.3751  | -4.8868  | -12.7294 | -6.4338  | -2.7424  | 2.6463   | -6.6974  | 16.0407 | -6.7270        | 3.12223   | 97.008                                     | 18933.11        |
| 119 | -7.0878  | -4.2634  | -12.2643 | -7.9914  | -2.2135  | 2.8975   | -7.2203  | 17.1381 | -6.8811        | 3.12223   | 100.066                                    | 22088.70        |
| 120 | -7.1129  | -1.0806  | -12.0159 | -8.6363  | 0.3207   | 2.6866   | -7.2406  | 15.5756 | -7.0660        | 3.12223   | 103.800                                    | 26576.93        |
| 121 | -6.8195  | -0.9315  | -11.2516 | -8.6973  | 0.4311   | 2.3659   | -7.2547  | 17.0503 | -6.6367        | 3.12223   | 95.237                                     | 17297.72        |
| 122 | -7.0916  | 0.6221   | -12.3002 | -9.5127  | 1.9641   | 2.0384   | -7.8434  | 17.0472 | -7.3597        | 3.12223   | 109.871                                    | 35654.59        |
| 123 | -4.9629  | 1.3969   | -9.6885  | -6.4287  | 2.4733   | 2.5353   | -5.4649  | 15.7711 | -5.1307        | 3.12223   | 68.110                                     | 3829.50         |
| 124 | -3.9488  | 3.1830   | -8.3098  | -5.3919  | 3.5650   | 4.0163   | -4.4689  | 16.4634 | -3.9679        | 3.12223   | 50.271                                     | 1192.03         |
| 125 | -4.0989  | 3.8649   | -9.3899  | -7.8267  | 4.2939   | 2.6007   | -6.0352  | 14.8798 | -5.4298        | 3.12223   | 73.137                                     | 5167.72         |
| 126 | -4.0113  | 3.3184   | -10.0194 | -8.5960  | 4.3962   | 2.4549   | -6.2848  | 16.2941 | -5.7974        | 3.12223   | 79.560                                     | 7467.48         |
| 127 | -5.1620  | 3.5227   | -10.9758 | -11.1085 | 4.6466   | 1.3276   | -7.6767  | 20.3467 | -6.7904        | 3.12223   | 98.261                                     | 20173.42        |
| 128 | -5.8112  | 5.9503   | -12.6317 | -14.0794 | 6.7888   | -0.1858  | -9.4140  | 18.1557 | -8.7219        | 3.12223   | 140.284                                    | 139254.08       |
| 129 | -4.1912  | 7.9440   | -10.7691 | -12.8918 | 8.4359   | 1.4253   | -7.8982  | 22.6076 | -6.8813        | 3.12223   | 100.070                                    | 22093.20        |
| 130 | -1.5847  | 10.5150  | -6.6160  | -9.4086  | 10.8993  | 4.2774   | -5.0498  | 27.0027 | -3.2276        | 3.12223   | 40.320                                     | 565.04          |
| 131 | -4.3015  | 4.4107   | -10.2683 | -12.2810 | 5.6121   | 2.2423   | -8.1015  | 25.7190 | -6.2866        | 3.12223   | 88.526                                     | 12185.21        |
| 132 | -9.2095  | -5.1171  | -16.1424 | -16.8592 | -3.2253  | -3.2119  | -13.0720 | 14.2318 | -12.1494       | 3.12223   | 233.222                                    | 4289176.28      |
| 133 | -13.8888 | -9.9142  | -19.6827 | -22.0600 | -8.2586  | -8.6493  | -18.0859 | 11.2298 | -16.4347       | 3.12223   | 382.473                                    | 311504697.68    |
| 134 | -13.4538 | -10.5159 | -19.8033 | -21.8414 | -9.9820  | -9.8119  | -17.9057 | 12.6634 | -16.2988       | 3.12223   | 377.175                                    | 271911698.98    |
| 135 | -13.5437 | -11.8570 | -20.0461 | -21.0817 | -11.4050 | -10.5032 | -17.6273 | 16.2938 | -15.8945       | 3.12223   | 361.635                                    | 181490523.75    |
| 136 | -10.1349 | -10.7733 | -16.2718 | -16.4271 | -9.9701  | -6.8750  | -13.7820 | 25.7842 | -11.4438       | 3.12223   | 212.171                                    | 2118181.82      |
| 137 | -7.1180  | -10.0555 | -14.2181 | -13.7471 | -9.0189  | -4.2856  | -11.0474 | 35.7338 | -8.3744        | 3.12223   | 132.173                                    | 98371.60        |
| 138 | -5.3329  | -7.6757  | -12.1611 | -11.2718 | -6.8013  | -1.6135  | -8.7042  | 35.9381 | -6.3276        | 3.12223   | 89.300                                     | 12695.85        |
| 139 | -4.3340  | -8.1189  | -11.2925 | -9.8345  | -7.1581  | -0.5860  | -7.6008  | 36.3041 | -5.3227        | 3.12223   | 71.317                                     | 4641.89         |
| 140 | -7.1687  | -9.1517  | -13.4621 | -14.6135 | -8.3047  | -1.7564  | -11.3416 | 37.5911 | -8.0685        | 3.12223   | 125.233                                    | 72445.25        |
| 141 | -8.5913  | -8.7243  | -14.0454 | -15.4598 | -7.8994  | -1.7698  | -11.8193 | 36.5557 | -8.7802        | 3.12223   | 141.668                                    | 147615.00       |
| 142 | -10.7362 | -10.3070 | -15.4924 | -16.6255 | -8.9924  | -2.9691  | -13.3949 | 35.6808 | -10.1631       | 3.12223   | 176.501                                    | 588514.23       |
| 143 | -9.0049  | -9.3809  | -13.0801 | -14.2856 | -8.0511  | -1.5051  | -11.3555 | 33.8053 | -8.2949        | 3.12223   | 130.352                                    | 90855.43        |
| 144 | -5.7075  | -8.4857  | -10.3577 | -10.1352 | -7.4125  | 1.5535   | -7.7759  | 26.6562 | -5.9165        | 3.12223   | 81.699                                     | 8413.26         |
| 145 | -5.9100  | -8.3683  | -9.4951  | -8.7342  | -7.2889  | 3.7003   | -6.9518  | 32.3097 | -4.5383        | 3.12223   | 58.684                                     | 2114.29         |
| 146 | -5.7840  | -9.4614  | -10.0991 | -7.2226  | -7.6122  | 4.3356   | -6.0213  | 29.9113 | -4.4558        | 3.12223   | 57.426                                     | 1946.13         |

| Obs | r1      | r2       | r3       | r4       | r5       | r6      | r7      | r8      | y <sub>i</sub> | $\hat{y}$ | $(y_i - \hat{y})^2$ | L( $\delta_i$ ) |
|-----|---------|----------|----------|----------|----------|---------|---------|---------|----------------|-----------|---------------------|-----------------|
| 147 | -7.1251 | -10.0552 | -11.6871 | -8.8555  | -8.1272  | 4.3788  | -7.3930 | 27.8521 | -5.9768        | 3.12223   | 82.792              | 8936.40         |
| 148 | -6.6375 | -11.2826 | -11.3400 | -8.4552  | -9.0507  | 3.2997  | -7.1664 | 27.3213 | -5.7690        | 3.12223   | 79.054              | 7258.23         |
| 149 | -8.2353 | -14.6507 | -13.9548 | -10.8868 | -12.0233 | 2.0976  | -9.1194 | 28.2722 | -7.8204        | 3.12223   | 119.741             | 56522.33        |
| 150 | -6.4854 | -13.9497 | -12.4551 | -7.5584  | -11.4472 | 2.9830  | -6.8177 | 21.3686 | -6.4871        | 3.12223   | 92.339              | 14892.02        |
| 151 | -6.9971 | -12.2608 | -11.7099 | -8.3634  | -9.9795  | 1.5092  | -7.1402 | 20.4113 | -6.6020        | 3.12223   | 94.561              | 16707.14        |
| 152 | -6.9142 | -12.4894 | -11.1832 | -8.3913  | -10.3154 | 3.3620  | -7.2463 | 25.0344 | -5.9309        | 3.12223   | 81.959              | 8535.03         |
| 153 | -8.0592 | -11.9434 | -12.1029 | -9.4419  | -9.8007  | 1.9853  | -8.2562 | 24.8939 | -6.8467        | 3.12223   | 99.379              | 21341.40        |
| 154 | -7.2615 | -10.6215 | -10.0548 | -6.5308  | -8.7091  | 3.4399  | -6.4445 | 29.1518 | -4.5330        | 3.12223   | 58.603              | 2103.05         |
| 155 | -7.7410 | -9.2552  | -9.1403  | -5.3959  | -7.3482  | 3.0777  | -5.9929 | 27.2116 | -4.0163        | 3.12223   | 50.959              | 1251.47         |
| 156 | -7.6905 | -8.1750  | -8.7487  | -5.9631  | -6.4430  | 2.3859  | -6.1336 | 26.7462 | -4.0782        | 3.12223   | 51.846              | 1331.82         |
| 157 | -5.9477 | -7.7418  | -6.9685  | -3.2636  | -6.0006  | 2.5743  | -4.0047 | 17.4484 | -3.0596        | 3.12223   | 38.215              | 476.69          |
| 158 | -5.3366 | -3.1262  | -5.3736  | -2.4952  | -2.6224  | 3.5343  | -3.0826 | 16.6157 | -2.0431        | 3.12223   | 26.681              | 168.93          |
| 159 | -4.3898 | -1.7640  | -3.1733  | 0.9308   | -0.7229  | 6.8628  | -0.8748 | 13.6225 | -0.0310        | 3.12223   | 9.943               | 19.26           |
| 160 | -5.7537 | -4.0050  | -5.1816  | -0.6574  | -3.0121  | 5.5182  | -2.0984 | 12.2662 | -1.7054        | 3.12223   | 23.306              | 119.08          |
| 161 | -4.7808 | -5.3869  | -4.7532  | -0.7577  | -4.4915  | 4.2256  | -1.6597 | 9.5007  | -1.7412        | 3.12223   | 23.653              | 123.60          |
| 162 | -4.9558 | -5.1128  | -4.9897  | -2.2209  | -4.2859  | 2.8235  | -2.4995 | 12.4209 | -2.1345        | 3.12223   | 27.633              | 185.60          |
| 163 | -4.0524 | -5.8486  | -4.2253  | -1.1125  | -4.5506  | 4.4776  | -1.8389 | 13.7647 | -1.2112        | 3.12223   | 18.778              | 70.87           |
| 164 | -5.6797 | -7.5556  | -6.5842  | -2.9710  | -5.8617  | 4.1217  | -3.5391 | 14.9885 | -2.9040        | 3.12223   | 36.315              | 407.11          |
| 165 | -5.8780 | -10.9348 | -6.5086  | -2.6787  | -9.0196  | 3.1669  | -3.5834 | 14.9611 | -2.8950        | 3.12223   | 36.207              | 403.41          |
| 166 | -6.2361 | -8.6861  | -6.2161  | -3.1519  | -7.1922  | 3.0324  | -3.9245 | 18.0153 | -2.6848        | 3.12223   | 33.722              | 325.82          |
| 167 | -5.6745 | -7.1453  | -5.5260  | -2.8169  | -5.5804  | 3.6847  | -3.2537 | 16.3549 | -2.2981        | 3.12223   | 29.379              | 219.52          |
| 168 | -5.5208 | -8.9401  | -6.0934  | -3.2432  | -7.0750  | 3.0889  | -3.6960 | 19.1245 | -2.4811        | 3.12223   | 31.397              | 264.72          |
| 169 | -5.6205 | -9.1717  | -6.7421  | -3.4901  | -7.0576  | 2.7827  | -3.7005 | 19.3035 | -2.8141        | 3.12223   | 35.240              | 371.61          |
| 170 | -5.4307 | -10.8805 | -7.0468  | -4.5959  | -8.5627  | 2.6009  | -4.3771 | 18.9095 | -3.3559        | 3.12223   | 41.966              | 643.26          |
| 171 | -4.7703 | -10.1538 | -5.9541  | -3.1052  | -7.8140  | 2.9749  | -3.6676 | 20.8460 | -2.1945        | 3.12223   | 28.268              | 197.40          |
| 172 | -2.6826 | -9.2727  | -2.6246  | -0.5679  | -6.3090  | 3.8870  | -1.3241 | 27.3453 | 0.8663         | 3.12223   | 5.089               | 6.29            |
| 173 | -2.1132 | -8.7604  | -0.4798  | 1.2434   | -5.9364  | 5.3829  | -0.1961 | 30.6599 | 2.7385         | 3.12223   | 0.147               | 0.08            |
| 174 | -2.4751 | -7.2858  | -0.5806  | 2.5904   | -4.4857  | 5.7308  | 0.2151  | 34.5534 | 3.4466         | 3.12223   | 0.105               | 0.05            |
| 175 | -1.6999 | -7.0278  | -0.8791  | 3.2555   | -4.2910  | 5.3704  | 0.7744  | 30.7639 | 3.2878         | 3.12223   | 0.027               | 0.01            |
| 176 | -1.5199 | -6.0057  | 0.6719   | 5.7867   | -3.3887  | 6.4744  | 2.3111  | 32.3038 | 4.9666         | 3.12223   | 3.402               | 1.00            |
| 177 | -0.9067 | -5.0606  | 1.2433   | 6.4677   | -2.2204  | 7.9569  | 2.7857  | 34.0474 | 5.7016         | 3.12223   | 6.653               | 1.66            |
| 178 | -0.4210 | -1.4335  | 2.3901   | 7.3758   | 1.5170   | 8.7077  | 3.2940  | 31.8068 | 6.3765         | 3.12223   | 10.590              | 2.29            |
| 179 | 0.6108  | 0.9582   | 4.3480   | 8.8463   | 4.3108   | 10.2342 | 4.7324  | 35.4968 | 8.2114         | 3.12223   | 25.900              | 4.10            |
| 180 | -0.7960 | 4.0691   | 2.3033   | 6.9790   | 6.3638   | 8.0327  | 3.8462  | 34.6017 | 6.5513         | 3.12223   | 11.758              | 2.46            |
| 181 | -0.1921 | 4.0676   | 3.8724   | 9.0869   | 6.4931   | 7.4084  | 5.1187  | 31.4492 | 7.6382         | 3.12223   | 20.394              | 3.53            |
| 182 | -0.1019 | 3.2238   | 4.6895   | 9.8464   | 5.7092   | 7.6333  | 5.4447  | 30.7001 | 8.1471         | 3.12223   | 25.249              | 4.03            |
| 183 | 1.2143  | 1.7498   | 6.1685   | 12.5613  | 4.5333   | 9.7782  | 7.3305  | 34.3072 | 10.1680        | 3.12223   | 49.643              | 6.05            |

| Obs | r1      | r2      | r3      | r4      | r5      | r6      | r7      | r8      | y <sub>i</sub> | $\hat{y}$ | (y <sub>i</sub> - $\hat{y}$ ) <sup>2</sup> | L( $\delta_i$ ) |
|-----|---------|---------|---------|---------|---------|---------|---------|---------|----------------|-----------|--------------------------------------------|-----------------|
| 184 | 2.2213  | 2.2963  | 6.7897  | 12.3680 | 5.0355  | 9.0229  | 7.6722  | 29.9176 | 10.0779        | 3.12223   | 48.381                                     | 5.96            |
| 185 | 2.3675  | 0.0208  | 5.7732  | 11.1331 | 2.5683  | 7.2949  | 7.1607  | 27.9379 | 9.0261         | 3.12223   | 34.856                                     | 4.91            |
| 186 | 5.2440  | -0.6287 | 9.7858  | 15.1053 | 1.3730  | 8.5743  | 10.0751 | 22.4082 | 11.8175        | 3.12223   | 75.608                                     | 7.70            |
| 187 | 4.7446  | -1.5953 | 7.9994  | 14.0926 | 0.3286  | 9.2773  | 9.2785  | 21.1690 | 10.6077        | 3.12223   | 56.032                                     | 6.49            |
| 188 | 4.0358  | -4.1143 | 6.9154  | 13.4173 | -2.0347 | 7.0401  | 8.4557  | 20.7767 | 9.6908         | 3.12223   | 43.147                                     | 5.57            |
| 189 | 5.0468  | -4.7398 | 7.0241  | 15.3926 | -2.7525 | 7.1747  | 9.5501  | 14.4207 | 9.9104         | 3.12223   | 46.080                                     | 5.79            |
| 190 | 5.4693  | -5.6421 | 7.2163  | 15.7316 | -3.5893 | 6.7428  | 9.8319  | 15.8554 | 10.2415        | 3.12223   | 50.683                                     | 6.12            |
| 191 | 5.5592  | -6.7775 | 6.7796  | 14.5211 | -4.4479 | 7.1530  | 9.6876  | 18.6676 | 9.9828         | 3.12223   | 47.068                                     | 5.86            |
| 192 | 7.0477  | -5.6692 | 9.0003  | 15.6570 | -3.5571 | 7.4878  | 10.4440 | 20.5283 | 11.5682        | 3.12223   | 71.335                                     | 7.45            |
| 193 | 5.6194  | -4.6352 | 7.9760  | 13.5386 | -2.6220 | 8.0093  | 9.2405  | 19.8561 | 10.3153        | 3.12223   | 51.741                                     | 6.19            |
| 194 | 4.0490  | -7.1256 | 6.3401  | 12.0185 | -5.2140 | 5.8410  | 7.6484  | 17.0979 | 8.5917         | 3.12223   | 29.915                                     | 4.47            |
| 195 | 4.4291  | -6.7267 | 7.6095  | 14.1268 | -4.8366 | 7.7210  | 9.1904  | 15.7369 | 9.8316         | 3.12223   | 45.016                                     | 5.71            |
| 196 | 4.9444  | -4.2492 | 8.8325  | 15.1641 | -2.9749 | 8.5341  | 9.8835  | 15.9332 | 10.7963        | 3.12223   | 58.892                                     | 6.67            |
| 197 | 6.8455  | -4.2832 | 11.5767 | 18.0592 | -2.8062 | 10.1552 | 12.1854 | 16.4045 | 13.1893        | 3.12223   | 101.346                                    | 9.07            |
| 198 | 7.5997  | -3.4177 | 13.0945 | 20.7115 | -1.6323 | 11.7236 | 13.6438 | 21.6648 | 15.2639        | 3.12223   | 147.419                                    | 11.14           |
| 199 | 6.5448  | -4.3240 | 11.8016 | 18.7488 | -2.4446 | 10.5282 | 12.2461 | 20.1772 | 13.7904        | 3.12223   | 113.809                                    | 9.67            |
| 200 | 5.6995  | -5.7906 | 10.0278 | 19.4145 | -3.6569 | 10.4138 | 12.0291 | 17.1584 | 12.8804        | 3.12223   | 95.222                                     | 8.76            |
| 201 | 8.0119  | -0.3815 | 13.6453 | 22.5175 | 0.7671  | 12.5337 | 15.0396 | 19.4987 | 16.0378        | 3.12223   | 166.812                                    | 11.92           |
| 202 | 15.5108 | 9.3635  | 24.1426 | 31.7330 | 9.9582  | 21.8304 | 23.5186 | 29.7732 | 25.6490        | 3.12223   | 507.455                                    | 21.53           |
| 203 | 20.7933 | 14.7873 | 29.3709 | 39.1493 | 15.6203 | 27.0758 | 29.9942 | 35.0041 | 31.6130        | 3.12223   | 811.726                                    | 27.49           |
| 204 | 19.4254 | 13.2738 | 27.8500 | 37.2266 | 15.9658 | 27.3355 | 28.3758 | 34.0422 | 30.1064        | 3.12223   | 728.148                                    | 25.98           |
| 205 | 18.7707 | 14.4105 | 27.7978 | 34.3717 | 17.3550 | 27.8915 | 26.9435 | 32.2679 | 28.9878        | 3.12223   | 669.027                                    | 24.87           |
| 206 | 16.1621 | 13.7327 | 25.1667 | 29.2560 | 16.3118 | 24.3687 | 23.5026 | 24.9964 | 25.1849        | 3.12223   | 486.762                                    | 21.06           |
| 207 | 13.8055 | 14.1186 | 23.3102 | 27.5264 | 15.7752 | 23.6513 | 21.8320 | 15.3091 | 22.7316        | 3.12223   | 384.527                                    | 18.61           |
| 208 | 11.1845 | 10.8843 | 19.3953 | 24.6000 | 12.3203 | 19.4369 | 18.8598 | 15.0265 | 19.6032        | 3.12223   | 271.624                                    | 15.48           |
| 209 | 10.8881 | 10.5175 | 18.8000 | 24.2334 | 11.9526 | 17.0697 | 18.3674 | 10.0216 | 18.6526        | 3.12223   | 241.191                                    | 14.53           |
| 210 | 13.1339 | 11.6057 | 20.5687 | 29.3813 | 12.9416 | 17.2380 | 21.8957 | 7.9013  | 21.1535        | 3.12223   | 325.126                                    | 17.03           |
| 211 | 13.6363 | 9.7317  | 18.7938 | 28.0111 | 10.5895 | 15.3215 | 20.5648 | 6.9434  | 19.7794        | 3.12223   | 277.463                                    | 15.66           |
| 212 | 16.4373 | 11.1558 | 21.5334 | 31.7355 | 11.5215 | 15.1142 | 24.0317 | 7.0168  | 22.5070        | 3.12223   | 375.771                                    | 18.38           |
| 213 | 12.5258 | 7.7734  | 15.6965 | 26.0133 | 8.7862  | 13.0085 | 19.2988 | 8.4347  | 17.7876        | 3.12223   | 215.074                                    | 13.67           |
| 214 | 10.7298 | 5.6061  | 14.7199 | 23.3887 | 7.2549  | 11.3008 | 17.0932 | 14.7630 | 16.7603        | 3.12223   | 185.998                                    | 12.64           |
| 215 | 11.5614 | 4.1826  | 14.6160 | 22.7463 | 6.6041  | 10.5535 | 17.0778 | 15.4763 | 16.6280        | 3.12223   | 182.405                                    | 12.51           |
| 216 | 10.8237 | 3.8927  | 13.1154 | 20.4306 | 6.3525  | 10.1603 | 15.4925 | 20.9979 | 15.6297        | 3.12223   | 156.438                                    | 11.51           |
| 217 | 9.8575  | 2.7423  | 12.7898 | 21.2043 | 5.5712  | 9.8914  | 15.3910 | 23.6815 | 15.8286        | 3.12223   | 161.452                                    | 11.71           |
| 218 | 9.4643  | 3.1590  | 12.6046 | 21.4080 | 5.7871  | 10.7075 | 15.4298 | 24.0202 | 15.8486        | 3.12223   | 161.961                                    | 11.73           |
| 219 | 10.4809 | 6.0551  | 13.9764 | 23.5234 | 8.3179  | 11.9840 | 17.1320 | 21.3895 | 17.0936        | 3.12223   | 195.200                                    | 12.97           |
| 220 | 8.2173  | 5.4047  | 11.9401 | 19.0311 | 7.7589  | 10.9065 | 14.0854 | 27.8110 | 15.0592        | 3.12223   | 142.491                                    | 10.94           |

| Obs | r1      | r2     | r3      | r4      | r5      | r6      | r7      | r8      | y <sub>i</sub> | $\hat{y}$ | $(y_i - \hat{y})^2$ | L( $\delta_i$ ) |
|-----|---------|--------|---------|---------|---------|---------|---------|---------|----------------|-----------|---------------------|-----------------|
| 221 | 8.8161  | 5.2577 | 12.0605 | 19.3234 | 7.6587  | 11.7479 | 14.2272 | 29.6355 | 15.4390        | 3.12223   | 151.703             | 11.32           |
| 222 | 8.8328  | 5.9454 | 11.4187 | 19.3460 | 8.0162  | 9.5108  | 14.3153 | 25.9740 | 14.7995        | 3.12223   | 136.359             | 10.68           |
| 223 | 10.1537 | 6.3275 | 12.1762 | 20.3053 | 8.2779  | 10.8352 | 15.3205 | 22.4571 | 15.3286        | 3.12223   | 148.994             | 11.21           |
| 224 | 9.8925  | 5.6658 | 11.9443 | 20.5678 | 7.5420  | 11.2791 | 15.3991 | 22.6073 | 15.3171        | 3.12223   | 148.715             | 11.19           |
| 225 | 8.8631  | 4.6874 | 10.1133 | 17.2731 | 6.6253  | 10.5550 | 13.5721 | 21.3326 | 13.2612        | 3.12223   | 102.799             | 9.14            |
| 226 | 8.7902  | 3.8836 | 8.7970  | 14.8874 | 5.8992  | 10.1700 | 12.3971 | 21.5350 | 11.9486        | 3.12223   | 77.905              | 7.83            |
| 227 | 6.3963  | 3.2224 | 5.7618  | 11.8037 | 5.2065  | 9.6077  | 9.7930  | 29.4505 | 10.0802        | 3.12223   | 48.413              | 5.96            |
| 228 | 6.6952  | 1.8262 | 5.3274  | 10.7646 | 4.5241  | 8.3996  | 9.2313  | 30.8290 | 9.6444         | 3.12223   | 42.539              | 5.52            |
| 229 | 7.9142  | 2.6886 | 7.1016  | 10.2941 | 5.2163  | 8.9730  | 9.6379  | 37.9916 | 11.0303        | 3.12223   | 62.538              | 6.91            |
| 230 | 8.1725  | 3.8855 | 7.7508  | 10.3971 | 6.0060  | 8.7963  | 9.6170  | 37.4435 | 11.2944        | 3.12223   | 66.785              | 7.17            |
| 231 | 7.4355  | 3.9428 | 7.1030  | 10.2977 | 5.9307  | 8.2915  | 9.3185  | 37.7439 | 10.9227        | 3.12223   | 60.847              | 6.80            |
| 232 | 7.3338  | 6.0729 | 7.6310  | 11.2430 | 7.7564  | 8.8984  | 9.6896  | 34.9700 | 11.2277        | 3.12223   | 65.699              | 7.11            |
| 233 | 6.8067  | 7.7989 | 6.7763  | 10.0218 | 8.9002  | 7.3847  | 9.0588  | 33.7608 | 10.2994        | 3.12223   | 51.511              | 6.18            |
| 234 | 7.5695  | 9.1608 | 8.1512  | 9.4076  | 9.6602  | 6.5922  | 9.0838  | 32.4383 | 10.6183        | 3.12223   | 56.192              | 6.50            |
| 235 | 7.8293  | 9.6082 | 9.0389  | 10.6095 | 10.1564 | 7.7570  | 10.1333 | 37.7281 | 11.9495        | 3.12223   | 77.921              | 7.83            |
| 236 | 9.2050  | 9.2048 | 10.4985 | 12.4813 | 9.5349  | 8.4568  | 11.7656 | 34.2686 | 13.0520        | 3.12223   | 98.600              | 8.93            |
| 237 | 8.0354  | 6.4513 | 9.3419  | 11.0200 | 6.7142  | 8.0826  | 10.0769 | 36.1284 | 12.0264        | 3.12223   | 79.285              | 7.90            |
| 238 | 7.6958  | 6.3185 | 9.7084  | 11.4034 | 6.9970  | 8.9697  | 10.2248 | 38.3920 | 12.4975        | 3.12223   | 87.896              | 8.38            |
| 239 | 8.6189  | 7.4671 | 9.8497  | 11.4637 | 7.6514  | 9.4115  | 10.3728 | 37.4662 | 12.6087        | 3.12223   | 89.993              | 8.49            |
| 240 | 8.2897  | 8.5114 | 10.5791 | 12.9682 | 8.7447  | 10.1405 | 11.5113 | 40.3385 | 13.7003        | 3.12223   | 111.896             | 9.58            |
| 241 | 8.8964  | 8.2759 | 11.6256 | 12.6591 | 8.5958  | 10.1085 | 11.6758 | 36.9085 | 13.8013        | 3.12223   | 114.043             | 9.68            |
| 242 | 8.5987  | 8.4008 | 10.6561 | 11.6842 | 8.1071  | 10.0267 | 10.9713 | 35.9620 | 12.9624        | 3.12223   | 96.830              | 8.84            |
| 243 | 7.8820  | 7.7078 | 9.1401  | 11.8747 | 7.6836  | 9.4684  | 10.6595 | 34.0367 | 12.1152        | 3.12223   | 80.874              | 7.99            |
| 244 | 8.1867  | 8.1129 | 9.1135  | 12.9068 | 8.2872  | 9.4156  | 11.4658 | 31.6649 | 12.2931        | 3.12223   | 84.105              | 8.17            |
| 245 | 7.2918  | 9.1689 | 7.6334  | 10.6148 | 8.8495  | 9.3005  | 9.5761  | 32.7235 | 10.8983        | 3.12223   | 60.467              | 6.78            |
| 246 | 7.9572  | 7.3176 | 8.6825  | 12.0494 | 7.3800  | 9.4461  | 10.3052 | 31.1410 | 11.6847        | 3.12223   | 73.316              | 7.56            |
| 247 | 7.9488  | 5.6996 | 8.5302  | 12.6198 | 5.7668  | 9.4841  | 10.5000 | 35.4889 | 12.1596        | 3.12223   | 81.675              | 8.04            |
| 248 | 7.7148  | 3.6598 | 8.6142  | 12.8716 | 3.6686  | 9.1297  | 10.7583 | 42.2020 | 12.8204        | 3.12223   | 94.054              | 8.70            |
| 249 | 8.4756  | 3.6158 | 9.5684  | 12.5166 | 3.2837  | 9.6712  | 10.6881 | 42.1185 | 13.1728        | 3.12223   | 101.014             | 9.05            |
| 250 | 9.5418  | 3.8576 | 9.8942  | 12.9566 | 3.5007  | 10.5250 | 11.0435 | 35.8529 | 13.0350        | 3.12223   | 98.263              | 8.91            |
| 251 | 8.2388  | 4.0676 | 7.2651  | 11.0328 | 3.5463  | 9.1733  | 9.4141  | 35.4687 | 11.0920        | 3.12223   | 63.518              | 6.97            |
| 252 | 9.3022  | 4.5174 | 8.6845  | 12.0733 | 3.9702  | 10.1200 | 10.5381 | 38.6697 | 12.4740        | 3.12223   | 87.456              | 8.35            |
| 253 | 9.5703  | 3.0285 | 9.0358  | 13.3870 | 2.4162  | 10.7756 | 11.2628 | 42.0450 | 13.3730        | 3.12223   | 105.079             | 9.25            |
| 254 | 9.5353  | 1.4761 | 8.2778  | 13.6981 | 0.8789  | 11.4359 | 11.5654 | 42.3300 | 13.2173        | 3.12223   | 101.911             | 9.10            |
| 255 | 10.2314 | 3.7182 | 9.6081  | 13.8649 | 3.3693  | 13.1102 | 11.9963 | 46.1339 | 14.3235        | 3.12223   | 125.468             | 10.20           |
| 256 | 8.1541  | 4.4961 | 7.5594  | 12.4037 | 3.9830  | 12.2757 | 10.4630 | 45.2692 | 12.6724        | 3.12223   | 91.206              | 8.55            |
| 257 | 8.6196  | 4.9014 | 8.7949  | 11.4475 | 4.4669  | 11.9166 | 10.2890 | 47.5265 | 13.1170        | 3.12223   | 99.895              | 8.99            |

| Obs | r1     | r2      | r3      | r4      | r5      | r6      | r7     | r8      | y <sub>i</sub> | $\hat{y}$ | (y <sub>i</sub> - $\hat{y}$ ) <sup>2</sup> | L( $\delta_i$ ) |
|-----|--------|---------|---------|---------|---------|---------|--------|---------|----------------|-----------|--------------------------------------------|-----------------|
| 258 | 7.7590 | 5.3808  | 8.2325  | 11.9178 | 5.2189  | 12.7909 | 9.9964 | 48.0309 | 12.9995        | 3.12223   | 97.559                                     | 8.88            |
| 259 | 6.7633 | 3.8941  | 6.6977  | 10.7277 | 4.0149  | 11.1876 | 9.2560 | 50.2878 | 12.0200        | 3.12223   | 79.171                                     | 7.90            |
| 260 | 5.9450 | 4.5652  | 6.5095  | 10.2691 | 4.5421  | 11.2732 | 8.7955 | 51.8442 | 11.8517        | 3.12223   | 76.203                                     | 7.73            |
| 261 | 5.1299 | 4.3134  | 6.0460  | 11.5183 | 4.0758  | 10.0717 | 8.7116 | 42.7913 | 11.0851        | 3.12223   | 63.407                                     | 6.96            |
| 262 | 2.5177 | 0.1183  | 2.4433  | 7.8362  | 0.2720  | 7.0623  | 5.8577 | 42.9027 | 7.9667         | 3.12223   | 23.469                                     | 3.85            |
| 263 | 2.5769 | -1.5677 | 2.2448  | 7.9360  | -1.2713 | 5.8943  | 5.9914 | 45.1352 | 8.0634         | 3.12223   | 24.415                                     | 3.95            |
| 264 | 2.9250 | -0.2153 | 2.6901  | 7.3882  | 0.6241  | 8.3522  | 5.7099 | 48.7819 | 8.5212         | 3.12223   | 29.148                                     | 4.40            |
| 265 | 4.1308 | -0.3484 | 3.7142  | 8.1388  | 0.3172  | 9.1561  | 6.3425 | 54.5702 | 9.8342         | 3.12223   | 45.050                                     | 5.71            |
| 266 | 3.5167 | -3.2231 | 2.7953  | 7.6869  | -2.4217 | 8.8222  | 5.8182 | 57.9489 | 9.4848         | 3.12223   | 40.483                                     | 5.36            |
| 267 | 3.3567 | -2.0531 | 2.0210  | 7.9026  | -1.3593 | 8.3989  | 5.6358 | 47.8507 | 8.3138         | 3.12223   | 26.953                                     | 4.20            |
| 268 | 4.6022 | -2.5276 | 3.9998  | 10.1172 | -1.9095 | 9.1909  | 7.2536 | 48.4193 | 10.0583        | 3.12223   | 48.109                                     | 5.94            |
| 269 | 5.5405 | -1.5660 | 4.1813  | 10.4421 | -1.5582 | 9.3201  | 7.7383 | 34.8391 | 9.1784         | 3.12223   | 36.677                                     | 5.06            |
| 270 | 5.5886 | -0.3075 | 4.1961  | 7.7619  | -0.2890 | 8.4481  | 6.6330 | 37.0587 | 8.5194         | 3.12223   | 29.130                                     | 4.40            |
| 271 | 6.2412 | 2.2648  | 5.8729  | 9.6472  | 2.2559  | 9.3630  | 8.0391 | 41.1063 | 10.3275        | 3.12223   | 51.915                                     | 6.21            |
| 272 | 5.9083 | 3.0710  | 5.8284  | 10.4045 | 3.4033  | 9.1504  | 8.1408 | 39.8991 | 10.3931        | 3.12223   | 52.866                                     | 6.27            |
| 273 | 7.1258 | 2.9341  | 5.1902  | 11.4605 | 3.3252  | 9.4809  | 8.8886 | 31.1455 | 9.8484         | 3.12223   | 45.241                                     | 5.73            |
| 274 | 7.5924 | 3.2242  | 4.8543  | 11.8565 | 3.8986  | 9.8746  | 8.9936 | 30.8494 | 9.8608         | 3.12223   | 45.409                                     | 5.74            |
| 275 | 7.1737 | 3.2056  | 3.6978  | 12.8676 | 3.8185  | 9.7500  | 9.0771 | 30.6053 | 9.6167         | 3.12223   | 42.178                                     | 5.50            |
| 276 | 6.1460 | 2.4825  | 1.5530  | 11.8777 | 2.8513  | 8.4722  | 7.9652 | 29.2784 | 8.1084         | 3.12223   | 24.862                                     | 3.99            |
| 277 | 5.9826 | 1.5560  | 1.9974  | 11.2785 | 2.3115  | 6.3357  | 7.1329 | 32.5797 | 8.2110         | 3.12223   | 25.895                                     | 4.09            |
| 278 | 5.5663 | 0.7270  | 1.4587  | 11.6146 | 1.8161  | 7.1469  | 7.0988 | 30.6946 | 7.9062         | 3.12223   | 22.886                                     | 3.79            |
| 279 | 4.7767 | -0.5776 | 0.9193  | 9.0808  | 0.2546  | 6.7611  | 6.1313 | 28.0196 | 6.5772         | 3.12223   | 11.937                                     | 2.49            |
| 280 | 5.5476 | -0.7810 | 1.7690  | 8.9398  | 0.2358  | 7.5948  | 6.2727 | 28.4928 | 7.0290         | 3.12223   | 15.263                                     | 2.93            |
| 281 | 5.7064 | 0.3830  | 1.6717  | 7.9921  | 1.6407  | 8.6841  | 5.8571 | 34.3444 | 7.2682         | 3.12223   | 17.189                                     | 3.16            |
| 282 | 5.5543 | 1.4439  | 1.7928  | 6.1600  | 2.2686  | 8.5182  | 4.8012 | 34.0359 | 6.6723         | 3.12223   | 12.603                                     | 2.58            |
| 283 | 8.2917 | 4.0439  | 6.4561  | 11.8222 | 4.8173  | 11.1234 | 8.9795 | 39.0314 | 11.3193        | 3.12223   | 67.191                                     | 7.20            |
| 284 | 8.3718 | 6.3637  | 5.2391  | 10.4784 | 6.7539  | 11.8835 | 8.5329 | 38.8548 | 10.4681        | 3.12223   | 53.962                                     | 6.35            |
| 285 | 9.3161 | 6.8719  | 5.6181  | 10.3511 | 6.9455  | 12.0795 | 8.5899 | 37.9443 | 10.6002        | 3.12223   | 55.920                                     | 6.48            |
| 286 | 7.7091 | 6.5436  | 3.2910  | 8.0996  | 6.1471  | 10.5613 | 6.7228 | 32.7676 | 8.2088         | 3.12223   | 25.873                                     | 4.09            |
| 287 | 6.9952 | 6.2022  | 2.2933  | 5.6368  | 5.6532  | 8.7994  | 5.3333 | 25.4026 | 6.2196         | 3.12223   | 9.593                                      | 2.14            |
| 288 | 5.4540 | 6.2922  | 0.9588  | 5.1701  | 6.0586  | 7.1421  | 4.4637 | 25.9937 | 5.3262         | 3.12223   | 4.858                                      | 1.31            |
| 289 | 4.5599 | 6.1913  | 0.8488  | 3.4706  | 5.9037  | 5.5800  | 3.3039 | 29.5700 | 4.8740         | 3.12223   | 3.069                                      | 0.93            |
| 290 | 4.2103 | 5.0711  | 0.7935  | 3.8595  | 5.0941  | 5.7998  | 3.4098 | 28.7416 | 4.8643         | 3.12223   | 3.035                                      | 0.92            |
| 291 | 2.5967 | 4.7445  | -2.1856 | 0.9766  | 4.3410  | 3.5271  | 1.3814 | 22.6838 | 1.9175         | 3.12223   | 1.451                                      | 1.13            |
| 292 | 2.4414 | 3.6845  | -1.4096 | 0.4423  | 3.5119  | 3.4588  | 1.2552 | 22.7642 | 2.0522         | 3.12223   | 1.145                                      | 0.85            |
| 293 | 2.6951 | 2.8343  | 0.4908  | 3.4013  | 2.7138  | 4.0552  | 2.9316 | 25.7107 | 4.0884         | 3.12223   | 0.934                                      | 0.35            |
| 294 | 3.2830 | 3.6901  | 1.4424  | 3.8300  | 3.6398  | 4.5138  | 3.4528 | 30.2348 | 5.1134         | 3.12223   | 3.965                                      | 1.13            |



| Obs | r1     | r2      | r3     | r4      | r5      | r6     | r7     | r8      | y <sub>i</sub> | $\hat{y}$ | (y <sub>i</sub> - $\hat{y}$ ) <sup>2</sup> | L( $\delta_i$ ) |
|-----|--------|---------|--------|---------|---------|--------|--------|---------|----------------|-----------|--------------------------------------------|-----------------|
| 295 | 5.3098 | 4.7456  | 3.4345 | 7.4167  | 4.8027  | 6.3839 | 5.9943 | 36.1387 | 7.9225         | 3.12223   | 23.043                                     | 3.81            |
| 296 | 6.0482 | 4.9523  | 5.0939 | 10.1101 | 5.1424  | 8.1654 | 7.5016 | 41.3313 | 10.0640        | 3.12223   | 48.189                                     | 5.94            |
| 297 | 6.0389 | 5.9541  | 4.4277 | 9.1652  | 5.6526  | 8.3275 | 7.0400 | 40.6748 | 9.4507         | 3.12223   | 40.050                                     | 5.33            |
| 298 | 5.1410 | 3.8440  | 3.6355 | 9.6340  | 3.8573  | 7.8691 | 6.6115 | 34.4265 | 8.5509         | 3.12223   | 29.470                                     | 4.43            |
| 299 | 4.6353 | 2.8049  | 2.0098 | 8.5802  | 2.8898  | 7.0591 | 5.7839 | 33.4017 | 7.3491         | 3.12223   | 17.866                                     | 3.24            |
| 300 | 5.5334 | 4.5570  | 2.8035 | 8.9064  | 5.1595  | 7.9350 | 6.1891 | 31.5622 | 7.7662         | 3.12223   | 21.567                                     | 3.65            |
| 301 | 5.5071 | 4.0414  | 3.2273 | 9.3156  | 3.9375  | 8.1363 | 6.2010 | 32.8139 | 8.1549         | 3.12223   | 25.327                                     | 4.04            |
| 302 | 4.7736 | 1.9929  | 2.3024 | 8.5778  | 2.4687  | 7.3542 | 5.5703 | 29.6197 | 7.1244         | 3.12223   | 16.017                                     | 3.02            |
| 303 | 4.7521 | 0.9522  | 2.4984 | 9.3277  | 1.4286  | 7.7336 | 6.0960 | 29.1598 | 7.4252         | 3.12223   | 18.516                                     | 3.32            |
| 304 | 5.5948 | 1.9567  | 3.7806 | 10.9950 | 1.9561  | 9.0173 | 7.2967 | 28.1492 | 8.5626         | 3.12223   | 29.597                                     | 4.44            |
| 305 | 6.8975 | 3.6940  | 5.0167 | 8.9911  | 3.5722  | 9.5496 | 6.5036 | 28.6728 | 8.6203         | 3.12223   | 30.229                                     | 4.50            |
| 306 | 7.0268 | 2.6783  | 4.0951 | 7.4166  | 2.7909  | 8.4305 | 5.5199 | 31.7065 | 7.9358         | 3.12223   | 23.170                                     | 3.82            |
| 307 | 6.7936 | 2.7542  | 3.0267 | 6.5919  | 3.1659  | 8.1778 | 5.5480 | 29.9280 | 7.1090         | 3.12223   | 15.894                                     | 3.01            |
| 308 | 5.8937 | 1.1179  | 1.6406 | 5.0491  | 0.8975  | 6.0262 | 4.2640 | 26.4974 | 5.5191         | 3.12223   | 5.745                                      | 1.49            |
| 309 | 5.3337 | 0.3973  | 1.9466 | 4.0071  | 0.4327  | 4.9130 | 3.6611 | 27.9648 | 5.3267         | 3.12223   | 4.860                                      | 1.31            |
| 310 | 5.4416 | 0.2985  | 2.1040 | 4.8878  | 0.2467  | 4.7321 | 3.7162 | 27.2086 | 5.5648         | 3.12223   | 5.966                                      | 1.53            |
| 311 | 5.2608 | -0.2410 | 1.8257 | 5.2391  | -0.2274 | 4.9536 | 3.7707 | 27.1757 | 5.5386         | 3.12223   | 5.839                                      | 1.51            |
| 312 | 5.2063 | -0.3654 | 2.9005 | 5.9122  | 0.0847  | 5.5959 | 4.1987 | 30.7627 | 6.5318         | 3.12223   | 11.625                                     | 2.44            |
| 313 | 5.4235 | 1.2580  | 2.8575 | 5.0888  | 1.2634  | 6.2376 | 4.0402 | 30.8488 | 6.3447         | 3.12223   | 10.384                                     | 2.26            |
| 314 | 5.9543 | 1.3260  | 3.6851 | 5.4142  | 1.2207  | 6.5697 | 4.6181 | 30.5877 | 6.8631         | 3.12223   | 13.994                                     | 2.76            |
| 315 | 6.3507 | -0.6230 | 3.9454 | 6.7524  | -0.0607 | 6.7815 | 5.6547 | 33.6991 | 7.7337         | 3.12223   | 21.266                                     | 3.62            |
| 316 | 5.3013 | 0.1328  | 2.3488 | 4.4909  | 0.7320  | 6.1184 | 4.1875 | 28.5987 | 5.7760         | 3.12223   | 7.043                                      | 1.72            |
| 317 | 4.7549 | 2.0692  | 2.2720 | 4.0363  | 2.4438  | 5.7643 | 3.9690 | 27.6814 | 5.4843         | 3.12223   | 5.579                                      | 1.46            |
| 318 | 4.3771 | 2.3437  | 1.4199 | 2.7129  | 2.7167  | 5.3579 | 2.9554 | 19.9477 | 3.9551         | 3.12223   | 0.694                                      | 0.27            |
| 319 | 3.5386 | 1.5901  | 1.2100 | 2.2410  | 2.1738  | 4.6268 | 2.6895 | 14.6728 | 3.1518         | 3.12223   | 0.001                                      | 0.00            |
| 320 | 5.1050 | 2.4125  | 3.2345 | 3.9612  | 1.7013  | 6.5578 | 4.0744 | 23.2460 | 5.5371         | 3.12223   | 5.832                                      | 1.50            |
| 321 | 6.8552 | 0.9979  | 5.7501 | 5.1339  | 0.5730  | 8.4410 | 5.0965 | 26.8300 | 7.4847         | 3.12223   | 19.031                                     | 3.38            |
| 322 | 6.2974 | -1.3106 | 5.7751 | 5.3285  | -1.5927 | 6.8465 | 4.7659 | 23.6348 | 7.0968         | 3.12223   | 15.797                                     | 2.99            |
| 323 | 5.7277 | 0.2205  | 5.0547 | 3.4200  | 0.1200  | 6.3585 | 3.8354 | 22.2826 | 6.0274         | 3.12223   | 8.440                                      | 1.96            |
| 324 | 6.7103 | 4.6088  | 6.6123 | 3.8801  | 3.6888  | 7.2868 | 4.1656 | 26.9514 | 7.3831         | 3.12223   | 18.155                                     | 3.27            |
| 325 | 5.9100 | 3.5635  | 6.1661 | 4.4246  | 3.1629  | 7.2212 | 4.1746 | 27.5144 | 7.3228         | 3.12223   | 17.645                                     | 3.22            |
| 326 | 5.9908 | 4.6047  | 6.2734 | 3.4786  | 4.5249  | 8.5984 | 3.6685 | 35.3877 | 7.8142         | 3.12223   | 22.015                                     | 3.70            |
| 327 | 4.9072 | 6.0904  | 5.1327 | 2.2361  | 6.2331  | 8.2074 | 2.6853 | 30.1079 | 6.3682         | 3.12223   | 10.536                                     | 2.28            |
| 328 | 5.9753 | 6.0723  | 6.3852 | 2.4595  | 5.7250  | 7.7637 | 3.2771 | 28.3077 | 6.9073         | 3.12223   | 14.327                                     | 2.81            |
| 329 | 6.0244 | 7.7615  | 7.4125 | 2.6643  | 7.2667  | 8.5153 | 3.3602 | 28.5576 | 7.4580         | 3.12223   | 18.799                                     | 3.35            |
| 330 | 4.1501 | 7.2841  | 5.4148 | 0.2887  | 7.0681  | 7.0181 | 1.0563 | 33.2161 | 5.9515         | 3.12223   | 8.005                                      | 1.89            |
| 331 | 4.9158 | 8.3295  | 6.6354 | 0.1088  | 8.0898  | 8.1403 | 1.3220 | 39.7249 | 7.1076         | 3.12223   | 15.884                                     | 3.00            |

| Obs | r1      | r2      | r3      | r4      | r5      | r6      | r7      | r8      | y <sub>i</sub> | $\hat{y}$ | (y <sub>i</sub> - $\hat{y}$ ) <sup>2</sup> | L( $\delta_i$ ) |
|-----|---------|---------|---------|---------|---------|---------|---------|---------|----------------|-----------|--------------------------------------------|-----------------|
| 332 | 8.2055  | 13.4637 | 10.9925 | 4.9256  | 12.6867 | 11.9913 | 5.2336  | 40.4496 | 11.1237        | 3.12223   | 64.023                                     | 7.00            |
| 333 | 7.1438  | 14.4233 | 8.5816  | 3.6454  | 13.3841 | 12.6036 | 4.1505  | 42.1771 | 9.7920         | 3.12223   | 44.485                                     | 5.67            |
| 334 | 8.1088  | 14.7972 | 9.3398  | 5.4522  | 14.4721 | 12.7935 | 5.5915  | 43.4556 | 10.9445        | 3.12223   | 61.188                                     | 6.82            |
| 335 | 6.1600  | 14.4409 | 7.5059  | 5.2893  | 14.2029 | 11.7678 | 5.2437  | 36.5726 | 9.3286         | 3.12223   | 38.519                                     | 5.21            |
| 336 | 6.8157  | 16.7699 | 8.1382  | 6.1006  | 16.8570 | 12.3453 | 5.6031  | 37.6393 | 10.0341        | 3.12223   | 47.773                                     | 5.91            |
| 337 | 5.9632  | 17.0994 | 6.9968  | 3.7001  | 17.5353 | 11.5172 | 4.0438  | 41.4058 | 8.9888         | 3.12223   | 34.417                                     | 4.87            |
| 338 | 4.6359  | 15.9756 | 5.0240  | 1.8709  | 16.2032 | 9.9115  | 3.0408  | 23.1833 | 5.8246         | 3.12223   | 7.303                                      | 1.77            |
| 339 | 4.2523  | 15.2534 | 5.1868  | 1.7388  | 16.0225 | 10.4174 | 2.8609  | 32.7613 | 6.6513         | 3.12223   | 12.454                                     | 2.56            |
| 340 | 5.1037  | 15.4967 | 6.1305  | 1.8674  | 16.1336 | 11.2408 | 3.1005  | 39.2698 | 7.7610         | 3.12223   | 21.518                                     | 3.65            |
| 341 | 5.6407  | 17.5534 | 5.9631  | -0.7043 | 17.6307 | 10.3364 | 2.0695  | 34.0935 | 6.4669         | 3.12223   | 11.186                                     | 2.38            |
| 342 | 4.3100  | 16.5488 | 4.1440  | -3.2952 | 15.4315 | 8.9114  | 0.1190  | 31.9247 | 4.4581         | 3.12223   | 1.784                                      | 0.60            |
| 343 | 3.4959  | 18.3671 | 5.3566  | -4.0304 | 16.6104 | 9.0809  | -0.3974 | 41.8884 | 5.5210         | 3.12223   | 5.754                                      | 1.49            |
| 344 | 4.5343  | 20.8924 | 7.7776  | -3.8207 | 18.8935 | 11.7772 | 0.5114  | 46.6136 | 7.2819         | 3.12223   | 17.302                                     | 3.18            |
| 345 | 5.6196  | 21.5929 | 9.8264  | -3.0648 | 19.2860 | 13.0993 | 1.2496  | 49.2680 | 8.7668         | 3.12223   | 31.861                                     | 4.65            |
| 346 | 5.3179  | 22.4343 | 9.7568  | -2.9228 | 20.1924 | 14.5019 | 1.3383  | 50.5104 | 8.9368         | 3.12223   | 33.809                                     | 4.82            |
| 347 | 4.3783  | 22.4278 | 7.9397  | -4.3381 | 19.9473 | 13.9361 | 0.1435  | 43.7625 | 7.0073         | 3.12223   | 15.094                                     | 2.91            |
| 348 | 4.1336  | 22.8892 | 8.7840  | -5.3033 | 20.6296 | 13.0077 | -0.3822 | 46.5248 | 7.2186         | 3.12223   | 16.781                                     | 3.11            |
| 349 | 4.8910  | 22.1421 | 9.3533  | -4.0453 | 20.2755 | 13.9034 | 0.1299  | 48.9044 | 8.1352         | 3.12223   | 25.129                                     | 4.02            |
| 350 | 5.1609  | 22.3165 | 10.1452 | -2.6338 | 20.6046 | 15.1177 | 1.2797  | 54.0172 | 9.4796         | 3.12223   | 40.417                                     | 5.36            |
| 351 | 5.1511  | 24.4551 | 10.8439 | -1.5643 | 22.3394 | 14.6308 | 1.8718  | 51.3333 | 9.8840         | 3.12223   | 45.722                                     | 5.76            |
| 352 | 5.7267  | 22.7209 | 11.8479 | 0.1212  | 21.3438 | 16.0445 | 3.0897  | 52.6581 | 11.0744        | 3.12223   | 63.237                                     | 6.95            |
| 353 | 3.8290  | 22.8932 | 8.4496  | -2.9037 | 21.0076 | 14.8693 | 0.7276  | 53.7991 | 8.5371         | 3.12223   | 29.320                                     | 4.42            |
| 354 | 2.8875  | 19.5482 | 8.2013  | -2.9044 | 18.4005 | 13.5084 | 0.1258  | 50.2585 | 7.8937         | 3.12223   | 22.767                                     | 3.78            |
| 355 | 2.4806  | 19.2150 | 8.0562  | -0.9930 | 18.1455 | 14.2708 | 1.2523  | 52.9437 | 8.7046         | 3.12223   | 31.163                                     | 4.59            |
| 356 | 3.8520  | 19.2853 | 10.7058 | -0.3535 | 18.3631 | 14.5397 | 2.0077  | 50.6758 | 9.9466         | 3.12223   | 46.571                                     | 5.83            |
| 357 | 3.8745  | 21.8511 | 11.3550 | 0.8655  | 20.5768 | 16.1869 | 2.7813  | 63.7040 | 11.8536        | 3.12223   | 76.236                                     | 7.73            |
| 358 | 4.1949  | 23.7357 | 12.4252 | -0.2995 | 22.2151 | 14.6618 | 2.3792  | 58.3196 | 11.4428        | 3.12223   | 69.232                                     | 7.32            |
| 359 | 3.7315  | 24.6613 | 10.0066 | 0.0681  | 23.2216 | 10.7090 | 2.0702  | 44.6282 | 9.1596         | 3.12223   | 36.450                                     | 5.04            |
| 360 | 5.3353  | 24.2150 | 11.4532 | 1.4871  | 22.6943 | 12.8721 | 3.1520  | 46.1701 | 10.5831        | 3.12223   | 55.664                                     | 6.46            |
| 361 | 4.0416  | 25.9113 | 11.4916 | 3.0103  | 24.2796 | 11.8837 | 3.6230  | 44.5664 | 10.7989        | 3.12223   | 58.932                                     | 6.68            |
| 362 | 2.6811  | 27.1836 | 9.1749  | 3.0428  | 26.2265 | 11.2122 | 2.9077  | 44.0148 | 9.6296         | 3.12223   | 42.345                                     | 5.51            |
| 363 | 1.1007  | 27.1208 | 5.1519  | -0.6805 | 27.1619 | 8.4434  | 0.0805  | 37.2003 | 5.8685         | 3.12223   | 7.542                                      | 1.81            |
| 364 | 2.3799  | 25.6805 | 6.6965  | 1.3394  | 25.9168 | 11.1319 | 1.4979  | 42.0736 | 7.8052         | 3.12223   | 21.931                                     | 3.69            |
| 365 | -0.0217 | 25.1657 | 4.2177  | -0.8823 | 25.5559 | 9.3100  | -0.5643 | 32.9906 | 4.9221         | 3.12223   | 3.240                                      | 0.97            |
| 366 | -1.7133 | 27.2794 | 1.8952  | -3.0012 | 27.7122 | 5.7905  | -2.1900 | 26.4879 | 2.4239         | 3.12223   | 0.488                                      | 0.31            |
| 367 | -1.6426 | 26.0025 | 2.1528  | -3.2677 | 27.0294 | 4.6461  | -2.5886 | 18.8840 | 1.6931         | 3.12223   | 2.043                                      | 1.75            |
| 368 | -1.1775 | 27.4514 | 3.1177  | -3.0072 | 28.3796 | 4.9427  | -1.9443 | 22.2089 | 2.5792         | 3.12223   | 0.295                                      | 0.18            |

| Obs | r1      | r2      | r3      | r4      | r5      | r6     | r7      | r8      | y <sub>i</sub> | $\hat{y}$ | $(y_i - \hat{y})^2$ | L( $\delta_i$ ) |
|-----|---------|---------|---------|---------|---------|--------|---------|---------|----------------|-----------|---------------------|-----------------|
| 369 | -0.8362 | 28.2633 | 3.8761  | -2.6267 | 29.1957 | 6.1218 | -1.4629 | 21.0838 | 3.0250         | 3.12223   | 0.009               | 0.00            |
| 370 | -3.1449 | 27.6371 | 2.3929  | -4.0589 | 28.6984 | 4.0830 | -3.0095 | 14.9413 | 1.0656         | 3.12223   | 4.230               | 4.76            |
| 371 | -4.4657 | 27.4558 | 0.6393  | -5.9197 | 29.9890 | 1.6977 | -4.3811 | 3.5016  | -1.4881        | 3.12223   | 21.255              | 94.90           |
| 372 | -4.7100 | 31.1399 | -0.3530 | -6.9395 | 32.3509 | 0.7109 | -5.1803 | 4.5357  | -2.1776        | 3.12223   | 28.089              | 194.01          |